Capstone: Using DDPM-Graph NN for Plasma Physics

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Motivation

- Magnetohydrodynamics (MHD) is actively studied field, with important use cases in astrophysics, high energy density physics, and **fusion energy research**
- **Solving Maxwell-Navier Stokes equs for MHD often takes days to do,** effectively delimiting important experiment for Inertial Confinement Fusion (ICF) and Magnetic Confinement Fusion (MCF)
- Interested in **predicting plasma density profile** for disruption prediction in tokamaks….to aid in **plasma confinement control**

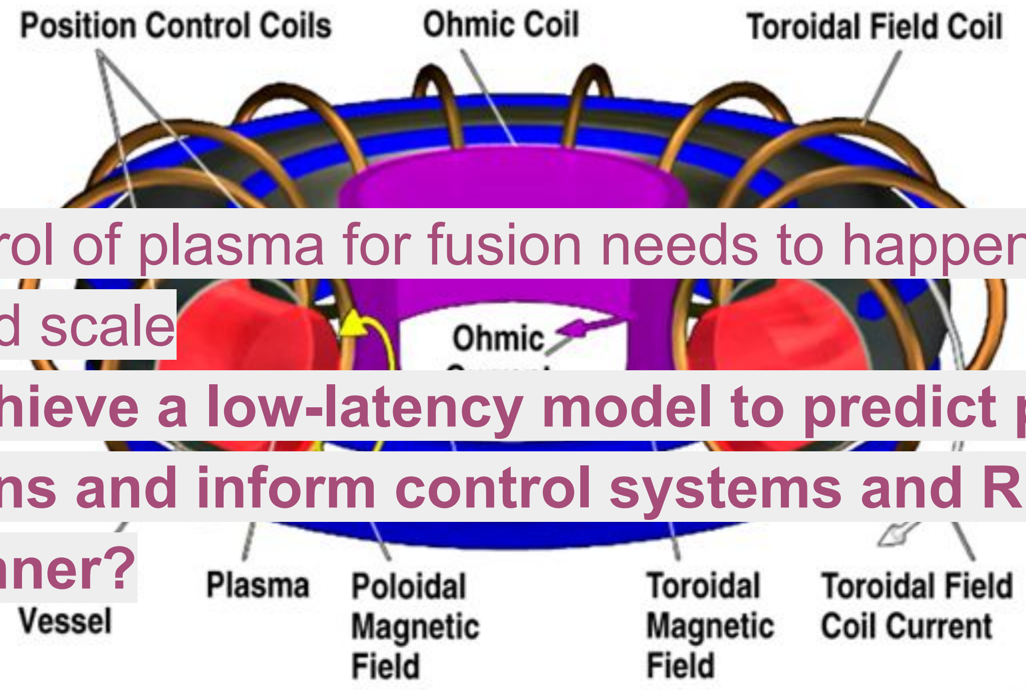
$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v} - \mathbf{B}\mathbf{B}) + \nabla p = 0$$

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{v} \times \mathbf{B}) = 0$$

[Fusion power: DeepMind uses AI to control plasma inside tokamak reactor | New Scientist]

Research Question



- Active control of plasma for fusion needs to happen at nanosecond scale
- **Can we achieve a low-latency model to predict plasma distributions and inform control systems and RL agents in timely manner?**

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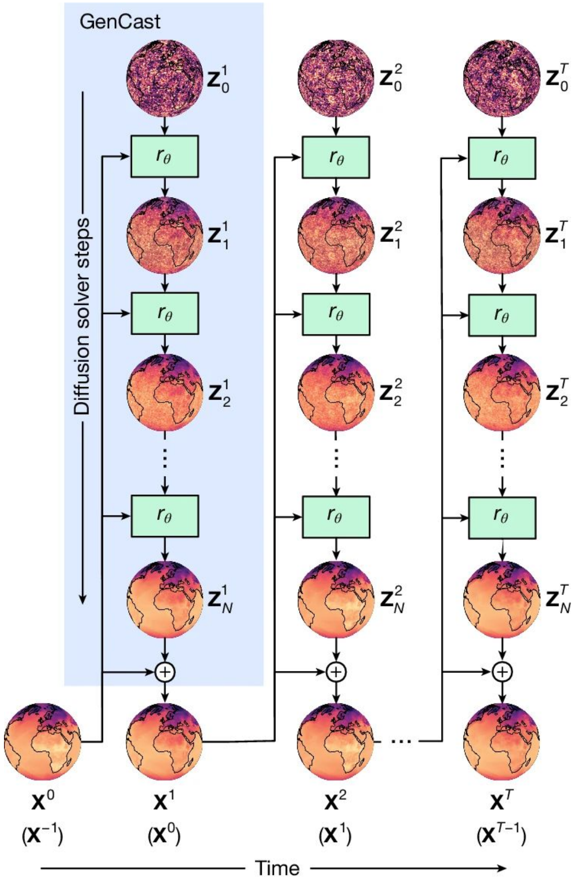
Case Studies…

- Physics-informed machine learning is a developing field already being applied to MHD and fusion energy research.
- Google and NVIDIA explored AI models that can predict daily weather patterns, on par with state of the art traditional models but much faster inference (2202.11214v1.pdf (FourCastNet neural operator model) and Probabilistic weather forecasting with machine learning | Nature (GenCast))
- Graph Attention models were explored for fluid dynamics with good MSE (Graph attention network-based fluid simulation model | AIP Advances | AIP Publishing)
- PINO-Diffusion hybrid for MHD modeling shows success (Resolving turbulent magnetohydrodynamics: a hybrid operator-diffusion framework - IOPscience)
- **Graph and Diffusion models have both demonstrated promise, how can we combine the strengths of both?**



Model Review

- Model is inspired by Google's use of graph diffusion in its weather prediction model, GenCast. Weather prediction has similar kinds of turbulence as plasmas. Too complicated and specific, had to build from ground up
- Novel approach -- **uses 3D Unet with a graph neural network at the bottleneck** to capture complex long-range, relational dependencies
- Encoder: $64^3 \rightarrow 32^3 \rightarrow 16^3$, Bottleneck processes 16^3 data with GATv2, Decoder 16^3 -> 32^3 -> 64^3
- Denoising diffusion adds noise and forces model to learn to predict relevant distributions
- **Predicts next frame of plasma from previous 3 frames (classifier free guidance conditioning)**
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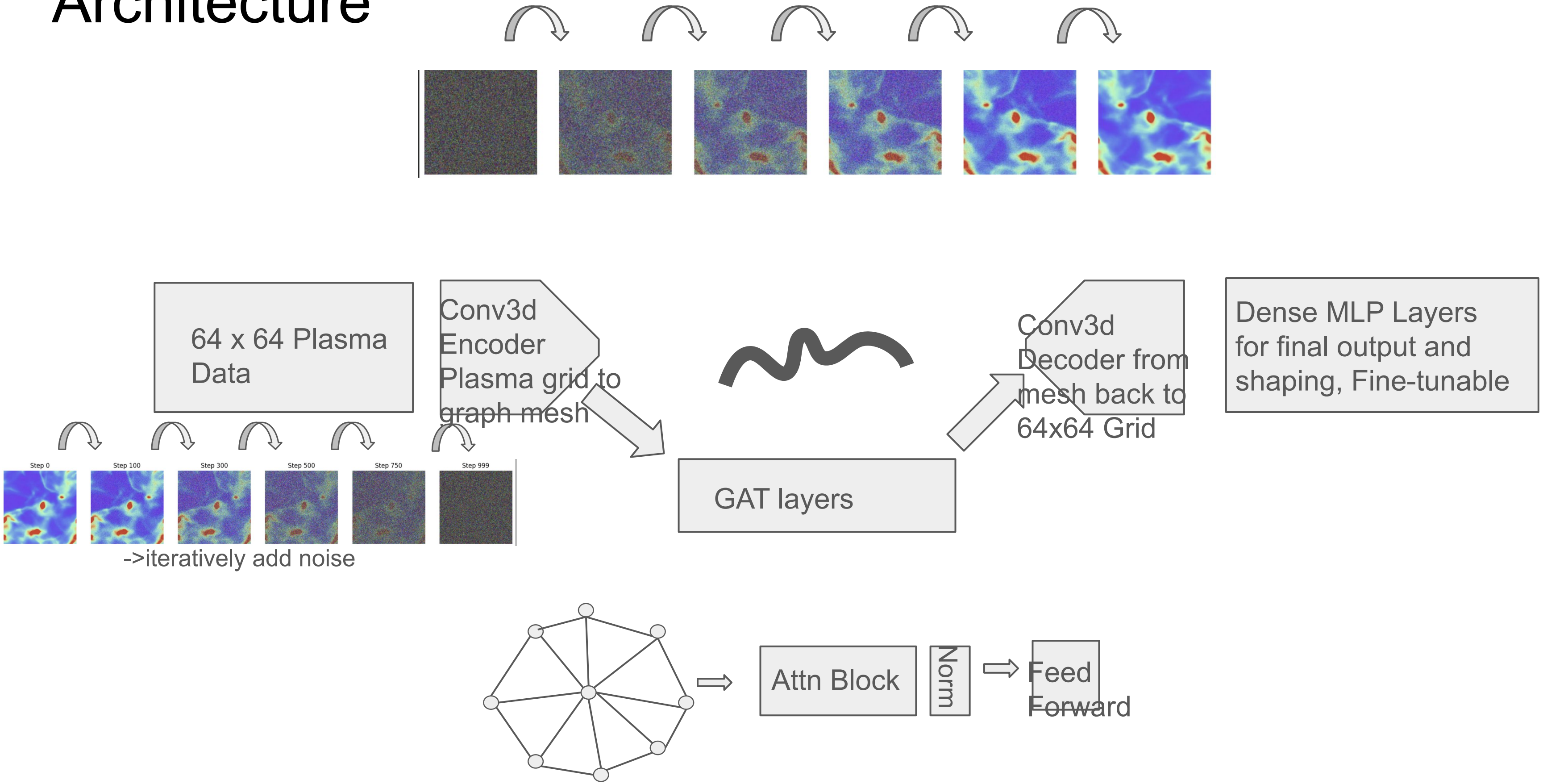
Architecture

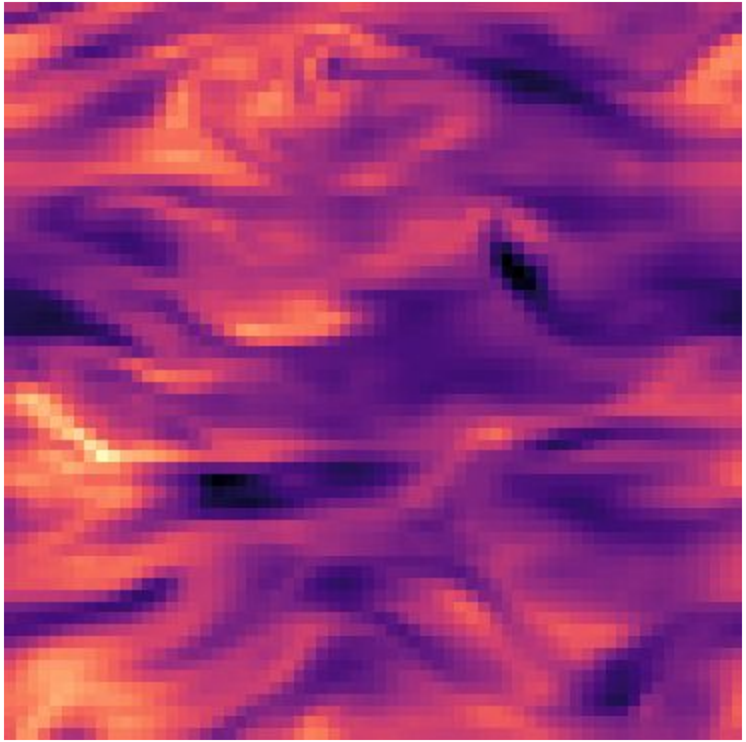


-Ref

[2212.12794v2.pdf](2212.12794v2.pdf)

Data Pipeline

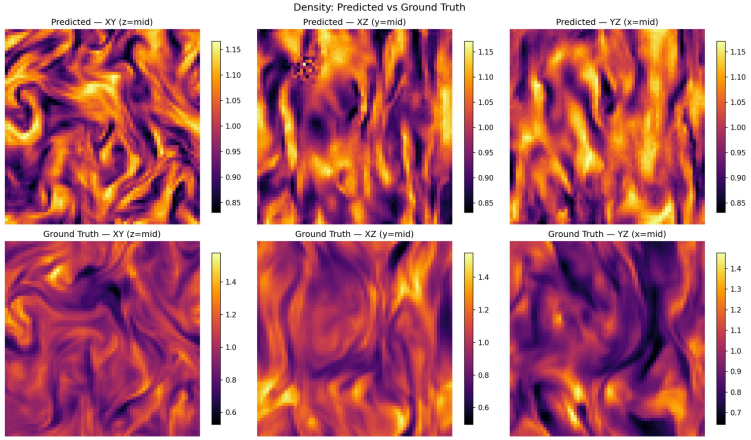
- Using Polymathic AI database for physics ML
  - 64 by 64 MHD h5 data, streamed from hugging face into Python
  - [polymathic-ai/MHD_64 · Datasets at Hugging Face](); Burkhart, B., Appel, S. M., Bialy, S., Cho, J., Christensen, A. J., Collins, D., Federrath, C., Fielding, D. B., Finkbeiner, D., Hill, A. S., et al. (2020). The catalogue for astrophysical turbulence simulations (CATS). *The Astrophysical Journal*, *905*(1), 14. https://doi.org/10.3847/1538-4357/abc484
  - Uniform grid, periodic boundary conditions, various compressibility conditions
  - Takes density, and magnetic field (3 comp vector) at every point
- Condition: [F_{t-3}, F_{t-2}, F_{t-1}], Target: [F_t]
- Sliding window sampling
- Per channel normalization before training

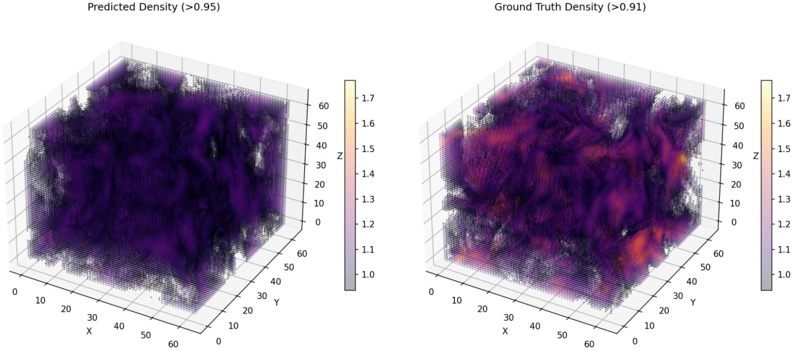


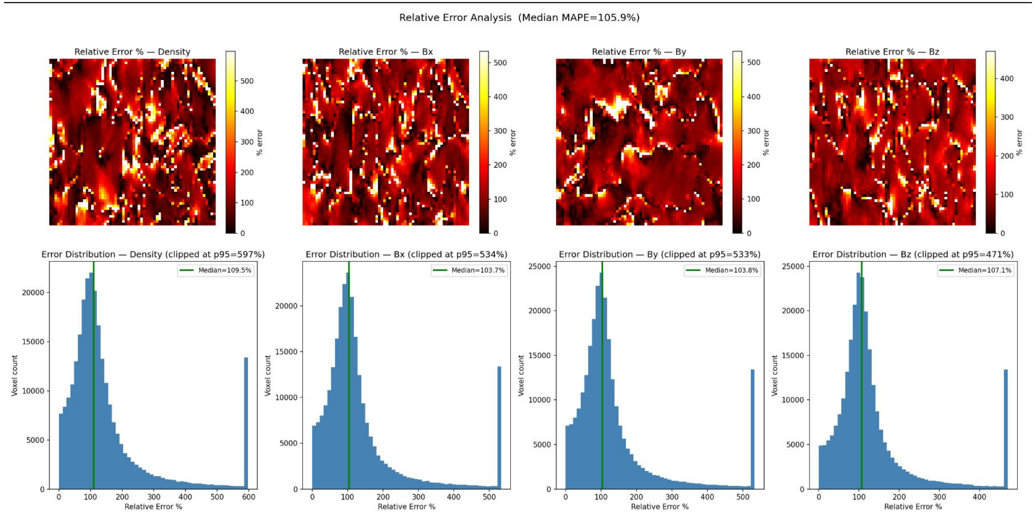
Training…

- Used local NVIDIA RTX 5070 GPU
- Trained for about 10 hours on 50 epochs, w/ each epoch containing 3,700 batches
- Gradient clipping + weighted SNR loss
- EMA
- Loss stabilizes at around 0.04 for both training and validation…suspiciously low value is due to SNR loss weighting (to account for variation in noise in scheduling)
- MSE and relative error loss tells completely different story….error off by multiple x the peak voxel intensity magnitude

Results (one example)…







(left) Random sample from validation set, looking at normalized color plot of midplane views and entire 3D volume. Can see that large-scale structure is captured but not fine detail
(top) [top row] Mean Abs Perc. Error (MAPE) of midplane of random sample; [bottom row] Distribution of relative error % within same sample, median error is >100% of target magnitude…

Next need to validate by going across all samples in set…

Baseline Models…

- The source hosting the data has baseline models
- Models include Tensorized Fourier Neural Operator, FNO, Unet, and ConvNext Unet
- They provide metric, VRMSE, which is not referenced elsewhere in literature
- Did not have time to validate myself…

Moving Forward…

- Model performs very unsatisfactorily at the moment….converges too quickly
- Learning entire 3D distributions is a very difficult challenge…2D predictions are already an active challenge in physics-informed physics
- Needs significantly better training and validation….way beyond 50 epochs and with more samples
- Conversation w/ LLNL scientist on physics informed diffusion models:
  - Investigate problems with normalization
  - Tune noise scheduler
  - Use physics informed classifier gradients in loss (Illuminating protein space with a programmable generative model | Nature)
- Will incorporate lessons during LLNL physics-informed model project this summer

Conclusion…

- **Model converges too quickly**, believe there is problem with SNR loss definition and normalization
- MSE and relative error are extremely high. **Model captures large-scale structure (visually) but not voxel-voxel detail**
- Need to incorporate more data and more effective training
- Make **model more physics informed** with elements like classifier gradients
- It is possible that scope of project turned out too ambitious in the end